

IMAT GK DRILL — 100 Questions

100 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · GK drill · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Reading Skills & Vocabulary in Context (Q1-8)

1. Read the following passage and answer the question.

For decades, adolescent risk-taking was attributed chiefly to ignorance: teenagers, the assumption went, simply failed to grasp how dangerous their behaviour was. Neuroimaging has complicated that picture. The limbic system, which processes reward, matures earlier than the prefrontal cortex, which supplies self-control, and the mismatch between the two is widest in mid-adolescence. Teenagers, it turns out, often appraise risks quite accurately; what sets them apart is that, especially under the gaze of peers, the anticipated reward looms larger than the anticipated harm.

Which of the following best captures the passage's main point?

- A) Adolescents take risks because they cannot understand danger at all.
- B) Uneven maturation of brain systems, rather than mere ignorance, helps explain adolescent risk-taking.
- C) Neuroimaging has disproved the very existence of adolescent risk-taking.
- D) Peer observation permanently damages the adolescent prefrontal cortex.
- E) The prefrontal cortex matures before the limbic system in most adolescents.

Answer: B

The passage explicitly retires the ignorance theory and replaces it with a developmental mismatch: the reward-processing limbic system matures before the self-controlling prefrontal cortex, the gap is widest in mid-adolescence, and peers amplify the pull of reward. The claim that teenagers cannot grasp danger at all is the very view the passage overturns, and it states the opposite order of maturation to the reversed version. Permanent brain damage and denial that the phenomenon exists appear nowhere.

2. For decades, adolescent risk-taking was attributed chiefly to ignorance: teenagers, the assumption went, simply failed to grasp how dangerous their behaviour was. Neuroimaging has complicated that picture. The limbic system, which processes reward, matures earlier than the prefrontal cortex, which supplies self-control, and the mismatch between the two is widest in mid-adolescence. Teenagers, it turns out, often appraise risks quite accurately; what sets them apart is that, especially under the gaze of peers, the anticipated reward looms larger than the anticipated harm.

In the passage, 'the anticipated reward looms larger than the anticipated harm' means that, for the adolescent, the reward:

- A) persists longer in memory.
- B) is approved of by peers.
- C) is physically larger in size.
- D) has been measured more precisely than the harm.
- E) is felt to matter more than the harm.

Answer: E

Looms larger is figurative: the expected gain carries more psychological weight than the expected loss, so for the adolescent the reward is felt to matter more than the harm. Nothing in the passage concerns physical size, duration in memory or measurement precision; peers only intensify the effect.

3. Read the following passage and answer the question.

Some researchers argue that reflecting a fraction of sunlight before it reaches the Earth's surface — so-called solar radiation management — could cool the planet quickly and cheaply, buying time while emissions fall. Critics reply that the technique is a sedative, not a cure: it eases the symptom of warming, temperature, while leaving its cause, greenhouse gases, untouched. They add a further worry: were the programme ever stopped abruptly, decades of masked warming could surface within a few years.

Which statement is supported by the passage?

- A) The technique eliminates the causes of climate change while ignoring its symptoms.
- B) Solar radiation management removes greenhouse gases from the atmosphere.
- C) Critics believe the programme could never be stopped for any reason.
- D) Solar radiation management could lower temperatures faster than emission cuts alone.
- E) All researchers agree the technique should be deployed immediately.

Answer: D

The researchers claim solar radiation management could cool the planet quickly and cheaply while buying time for emission cuts, so it would act on temperature faster than cuts alone. The critics say the opposite of removing greenhouse gases (the cause is left untouched), their worry is about stopping abruptly rather than an inability to stop, and no consensus on immediate deployment is claimed.

4. Some researchers argue that reflecting a fraction of sunlight before it reaches the Earth's surface — so-called solar radiation management — could cool the planet quickly and cheaply, buying time while emissions fall. Critics reply that the technique is a sedative, not a cure: it eases the symptom of warming, temperature, while leaving its cause, greenhouse gases, untouched. They add a further worry: were the programme ever stopped abruptly, decades of masked warming could surface within a few years.

The critics' phrase 'a sedative, not a cure' implies that solar radiation management:

- A) is prohibited under international law.
- B) has never been modelled by scientists.
- C) relieves a symptom of warming without removing its cause.
- D) costs more than conventional emission reductions.
- E) is medically dangerous to the researchers themselves.

Answer: C

A sedative dulls a symptom without curing the disease: here the symptom is temperature and the cause is greenhouse gases, so the metaphor says the technique relieves a symptom of warming without removing its cause. Medically dangerous, illegal, unmodelled and overpriced readings import ideas the passage never raises.

5. Read the following passage and answer the question.

Deep-learning systems now match or exceed specialists at reading certain retinal scans, and hospitals are eager to deploy them. Enthusiasm, however, can outrun evidence. An algorithm trained on patients from a handful of teaching hospitals may falter when it meets the cameras, the lighting and the patient mix of ordinary clinics. Its headline accuracy, measured on data that resemble its training set, is not guaranteed to travel. What a clinic needs is not a demonstration but a deployment study: performance measured where the machine will actually be used.

In the passage, 'is not guaranteed to travel' suggests that the algorithm's reported accuracy:

- A)** may fail to transfer to settings unlike those it was trained in.
- B)** depends on the price of the scanning equipment.
- C)** improves as more cameras are installed.
- D)** has been refuted by the specialists who trained it.
- E)** breaches patient-confidentiality rules across borders.

Answer: A

Travel is figurative: accuracy measured on data resembling the training set may not hold in ordinary clinics with different cameras, lighting and patients, so it may fail to transfer to settings unlike those it was trained in. The confidentiality, price, literal-journey and refutation readings are absent from the text.

6. Deep-learning systems now match or exceed specialists at reading certain retinal scans, and hospitals are eager to deploy them. Enthusiasm, however, can outrun evidence. An algorithm trained on patients from a handful of teaching hospitals may falter when it meets the cameras, the lighting and the patient mix of ordinary clinics. Its headline accuracy, measured on data that resemble its training set, is not guaranteed to travel. What a clinic needs is not a demonstration but a deployment study: performance measured where the machine will actually be used.

Which of the following best expresses the author's central claim?

- A)** Algorithms should be trained exclusively in ordinary clinics.
- B)** Teaching hospitals systematically exaggerate scanner costs.
- C)** Retinal specialists will soon become redundant.
- D)** Deep learning has no useful role in ophthalmology.
- E)** Adoption should rest on performance validated in real deployment settings, not on headline benchmarks.

Answer: E

The thesis sits in the final sentence: a clinic needs a deployment study, performance measured where the machine will actually be used, rather than a headline benchmark on training-like data. The passage endorses the technology, so its rejection or the redundancy of specialists overstates it; training exclusively in ordinary clinics distorts the proposal, and scanner costs never appear.

7. Read the following passage and answer the question.

The microbes of the human gut ferment dietary fibres that our own enzymes cannot break down, releasing short-chain fatty acids that feed the cells of the intestinal wall and help keep inflammation in check. A diet chronically poor in fibre therefore does more than deprive the body of roughage: it starves a metabolic partnership refined over millions of years. Deprived of fibre, some microbes turn to an alternative fare — the mucus that lines the gut — thinning the barrier that separates the microbial world from the body's own tissues.

Which conclusion is best supported by the passage?

- A) All gut microbes are harmful to human health.
- B) Short-chain fatty acids are produced by human digestive enzymes.
- C) Chronic fibre deficiency may weaken the barrier protecting gut tissues.
- D) Mucus is the preferred food of all gut bacteria.
- E) Inflammation is caused exclusively by dietary fibre.

Answer: C

Follow the chain: starved of fibre, some microbes eat the gut mucus instead, and that mucus is the barrier shielding the body's own tissues, so chronic fibre deficiency may weaken the barrier protecting gut tissues. The fatty acids come from microbes, not human enzymes, and the hedged wording supports no absolute claims about all microbes or exclusive causes.

8. The microbes of the human gut ferment dietary fibres that our own enzymes cannot break down, releasing short-chain fatty acids that feed the cells of the intestinal wall and help keep inflammation in check. A diet chronically poor in fibre therefore does more than deprive the body of roughage: it starves a metabolic partnership refined over millions of years. Deprived of fibre, some microbes turn to an alternative fare — the mucus that lines the gut — thinning the barrier that separates the microbial world from the body's own tissues.

In the passage, 'roughage' most nearly means:

- A) indigestible plant material.
- B) refined sugars.
- C) vitamins absorbed in the colon.
- D) coarse meat proteins.
- E) the mucus lining of the gut.

Answer: A

Roughage is the everyday name for dietary fibre, the indigestible plant material human enzymes cannot break down, exactly as the passage's opening clause defines it. Meat proteins, refined sugars, vitamins and the mucus lining do not fit the in-passage definition.

Section B - Italian Constitution & Institutions (Q9-24)

9. How many judges sit on Italy's Constitutional Court?

- A) 9 B) 12 C) 15 D) 21 E) 25

Answer: C

Italy's Constitutional Court has fifteen judges, each serving a single non-renewable nine-year term. The number nine belongs to the US Supreme Court and is the classic confusion here.

- 10.** The fifteen judges of the Constitutional Court are appointed by:
- A)** the President of the Republic alone.
 - B)** the President of the Republic (5), Parliament in joint session (5) and the supreme magistrature (5).
 - C)** direct popular election every nine years.
 - D)** the supreme courts alone.
 - E)** the Parliament alone, by two-thirds majority.

Answer: B

Article 135 splits the appointments into equal thirds: five judges named by the President of the Republic, five elected by Parliament in joint session and five by the supreme ordinary and administrative courts. Any single-body formula breaks the deliberate balance, and there is no popular election.

- 11.** The Italian Constitution was approved by the Constituent Assembly on 22 December 1947 and entered into force on:
- A)** 1 January 1948.
 - B)** 1 May 1949.
 - C)** 2 June 1946.
 - D)** 18 April 1948.
 - E)** 25 April 1947.

Answer: A

The sequence is fixed: referendum and Constituent Assembly on 2 June 1946, approval on 22 December 1947, entry into force on 1 January 1948. The April 1948 and 1949 dates fall after it, and the 1946 and 1947 dates earlier in the same chain.

- 12.** How many articles does the Italian Constitution contain?
- A)** 100 **B)** 114 **C)** 129 **D)** 139 **E)** 150

Answer: D

The constitutional text runs to 139 articles, organised into fundamental principles, Part I on rights and duties and Part II on the organisation of the Republic, plus transitional provisions. None of the smaller figures matches the official count of articles.

- 13.** The President of the Italian Republic is elected by:
- A)** the Constitutional Court.
 - B)** direct popular vote in two rounds.
 - C)** the Council of Ministers.
 - D)** the Chamber of Deputies alone.
 - E)** Parliament meeting in joint session, integrated with delegates of the regions.

Answer: E

The electoral college is both chambers in joint session plus three delegates per region (one only for Valle d'Aosta, fifty-eight in all); two thirds is required for the first three ballots, an absolute majority thereafter. The election is indirect, so direct popular voting describes other republics, not Italy.

14. Italy's Parliament consists of:

- A)** the Chamber of Deputies and the Regional Councils.
- B)** two chambers — the Chamber of Deputies and the Senate of the Republic — with identical powers.
- C)** the Senate and the Council of State.
- D)** a single National Assembly.
- E)** the Chamber of Deputies and representatives of the municipalities.

Answer: B

Italy has perfect bicameralism: the Chamber of Deputies and the Senate of the Republic sit for five years and hold identical powers, voting on the same bills. The Council of State is an administrative court, and regional councils or municipal representatives are not parliamentary chambers.

15. Under Article 94, the Government (Council of Ministers) must obtain the confidence of:

- A)** the Chamber of Deputies only.
- B)** the Constitutional Court.
- C)** the Senate only.
- D)** both the Chamber of Deputies and the Senate.
- E)** the President of the Republic.

Answer: D

Under Article 94 a new government must appear before both chambers and obtain their confidence within ten days of appointment; either house can withdraw it by a reasoned motion. Confidence is not a presidential, senatorial-only or judicial prerogative.

16. Following the 2020 constitutional cutback, first applied in the 2022 elections, the two chambers have:

- A)** 400 deputies and 200 senators.
- B)** 400 deputies and 100 senators.
- C)** 500 deputies and 250 senators.
- D)** 630 deputies and 315 senators.
- E)** 630 deputies and 200 senators.

Answer: A

Constitutional law 1 of 2020 cut Parliament to 400 deputies and 200 elected senators, first applied in 2022, nearly halving the previous 630 and 315. Any pair containing 630 or 315 belongs to the pre-reform chambers.

17. The Constitutional Court decides primarily on:

- A)** the legality of government contracts.
- B)** disputes between private citizens.
- C)** criminal appeals against convictions.
- D)** the constitutionality of national and regional laws.
- E)** the accession of new members to the European Union.

Answer: D

The Court's primary function is judging the constitutionality of national and regional laws. It also settles conflicts between organs of the state and between state and regions, and tries the President for crimes against the Constitution, but it never hears ordinary criminal appeals or private disputes.

18. Article 1 of the Constitution states that Italy is:

- A)** a constitutional monarchy.
- B)** a democratic Republic founded on labour.
- C)** a Republic founded on private property.
- D)** a presidential Republic.
- E)** a confederation of autonomous regions.

Answer: B

Article 1 reads: Italy is a democratic Republic founded on labour, with sovereignty belonging to the people who exercise it in the forms and limits of the Constitution. The monarchy fell in the 1946 referendum, and neither confederal, presidential nor property-based wording appears.

19. How many of Italy's twenty regions have a special statute and autonomy?

- A)** 2 **B)** 3 **C)** 4 **D)** 5 **E)** 10

Answer: D

Five regions hold special statutes and enlarged autonomy: Sicily, Sardinia, Valle d'Aosta, Trentino-Alto Adige/Südtirol and Friuli-Venezia Giulia. The remaining fifteen are ordinary-statute regions.

20. An abrogative referendum (Article 75) is valid when:

- A)** turnout exceeds seventy-five per cent of voters.
- B)** the President of the Republic authorises it.
- C)** a majority of valid votes approve it AND at least half the eligible electorate turns out.
- D)** a majority of the Chamber of Deputies passes it first.
- E)** one third of the Senate requests it.

Answer: C

An abrogative referendum succeeds when a majority of valid ballots favour repeal and turnout reaches at least half of those entitled to vote. It is called by 500,000 voters or five regional councils, and taxes, the budget, amnesties and ratified treaties are exempt; no presidential authorisation, prior Chamber vote or seventy-five per cent rule exists.

21. The President of the Council of Ministers (the prime minister) is:

- A)** nominated by the European Council.
- B)** the eldest senator, by automatic succession.
- C)** elected directly by the citizens.
- D)** chosen by the Constitutional Court.
- E)** appointed by the President of the Republic and must hold the confidence of Parliament.

Answer: E

The President of the Republic appoints the prime minister and, on the premier's proposal, the ministers, and the government must then win the confidence of Parliament in both chambers. Italy elects no prime minister directly, and neither the Constitutional Court, automatic seniority nor the European Council is involved.

22. Who has served as President of the Italian Republic since 2015 and was confirmed for a second term in 2022?

- A)** Sergio Mattarella
- B)** Carlo Azeglio Ciampi
- C)** Francesco Cossiga
- D)** Sandro Pertini
- E)** Giorgio Napolitano

Answer: A

Sergio Mattarella took office in 2015 and was re-elected in January 2022 for a second term running to 2029. Napolitano (2006 to 2015) was his predecessor, and Ciampi, Pertini and Cossiga sit earlier in the presidential list.

23. Since October 2022, the President of the Council of Ministers has been:

- A)** Matteo Renzi
- B)** Paolo Gentiloni
- C)** Mario Draghi
- D)** Enrico Letta
- E)** Giorgia Meloni

Answer: E

Giorgia Meloni leads the government formed in October 2022 after winning the September elections, Italy's first woman prime minister, having succeeded technocrat Mario Draghi. Gentiloni, Renzi and Letta are earlier premiers or party figures, not the incumbent.

24. Italy's national day, 2 June, commemorates:

- A)** the entry into force of the Constitution in 1948.
- B)** the 1946 referendum in which Italy chose to become a Republic.
- C)** the unification of 1861.
- D)** the March on Rome of 1922.
- E)** the liberation of 1945.

Answer: B

Festa della Repubblica recalls 2 June 1946, the institutional referendum in which Italians chose a Republic over the monarchy and elected the Constituent Assembly, the first national ballot open to women. Liberation Day falls on 25 April, unification on 17 March 1861, and the Constitution entered into force on 1 January 1948.

Section C - European Union (Q25-40)

25. The European Central Bank has its seat in:

- A)** Brussels.
- B)** The Hague.
- C)** Frankfurt am Main.
- D)** Paris.
- E)** Luxembourg.

Answer: C

The European Central Bank is seated in Frankfurt am Main, where it steers euro-area monetary policy. Brussels hosts the Commission's main offices and Luxembourg the Court of Justice and Court of Auditors; The Hague holds no major EU institution.

26. The official seat of the European Parliament — where the twelve monthly plenary sessions are held — is:

- A)** Geneva. **B)** Strasbourg. **C)** Bonn. **D)** Luxembourg. **E)** Brussels.

Answer: B

The twelve monthly plenary sessions of the European Parliament take place in Strasbourg, its official seat; Brussels hosts committee meetings and the political groups, Luxembourg the General Secretariat. Geneva is not an EU seat at all.

27. The Court of Justice of the European Union (CJEU) sits in:

- A)** Brussels. **B)** Vienna. **C)** Luxembourg. **D)** Strasbourg. **E)** Frankfurt.

Answer: C

The Court of Justice of the European Union sits in Luxembourg. The deliberate trap is the European Court of Human Rights, which sits in Strasbourg but belongs to the Council of Europe, an organisation outside the EU.

28. The European Commission is best described as:

- A)** the EU's executive arm: it proposes legislation and guards the treaties, with one member per member state.
B) a court hearing human-rights appeals.
C) the assembly of national parliament speakers.
D) a committee of central-bank governors.
E) the upper house of the EU legislature.

Answer: A

The Commission is the Union's executive: it holds the near-monopoly on proposing legislation, polices compliance as guardian of the treaties, and consists of one commissioner per member state under its president. It is neither a court, a parliament, a speakers' assembly nor a committee of central-bank governors.

29. The European Council consists of:

- A)** the EU commissioners.
B) the foreign ministers of the member states.
C) all members of the European Parliament.
D) the heads of state or government of the member states, with its President and the Commission President.
E) the presidents of the national constitutional courts.

Answer: D

The European Council unites the heads of state or government of the member states with its own president and the Commission president, setting the EU's general political direction from Brussels. The foreign ministers form the separate Council of the EU, which legislates alongside Parliament, a frequent confusion.

- 30.** The Treaty of Rome (1957) established:
- A)** the North Atlantic Treaty Organization.
 - B)** the Western European Union.
 - C)** the Council of Europe.
 - D)** the European Monetary System.
 - E)** the European Economic Community (EEC).

Answer: E

The Treaty of Rome of 25 March 1957 established the European Economic Community among six founders (Italy, France, West Germany, Belgium, the Netherlands, Luxembourg). Coal and Steel came from Paris 1951, NATO from Washington 1949, and the EU itself from Maastricht 1992.

- 31.** The Maastricht Treaty (1992) is notable for:
- A)** creating the Schengen area.
 - B)** founding the EEC.
 - C)** creating the Council of Europe.
 - D)** establishing the European Union and the path to the single currency.
 - E)** abolishing national parliaments.

Answer: D

Signed in 1992, Maastricht transformed the communities into the European Union and fixed the convergence criteria opening the path to the single currency. The EEC already existed (1957), the Council of Europe is older still and not an EU body, and Schengen is the 1985 agreement applied from 1995.

- 32.** Euro banknotes and coins entered circulation on:
- A)** 1 January 2002.
 - B)** 1 January 1999.
 - C)** 1 January 2000.
 - D)** 1 January 2004.
 - E)** 1 July 2013.

Answer: A

Notes and coins entered circulation on 1 January 2002 in twelve countries; the euro had existed since 1 January 1999 only as book money. The 1999 date is the standard trap.

- 33.** After the United Kingdom left the EU on 31 January 2020 (Brexit), the number of member states is:
- A)** 25 **B)** 26 **C)** 27 **D)** 28 **E)** 30

Answer: C

After the UK's withdrawal on 31 January 2020 the Union counts 27 member states, the first and only reduction in its history. Croatia, the 28th entrant of 2013, remains the most recent to join.

- 34.** Which country was the last to join the European Union, in 2013?
- A)** Romania **B)** Bulgaria **C)** Slovenia **D)** Hungary **E)** Croatia

Answer: E

Croatia joined on 1 July 2013 as the 28th member, the most recent accession to date. Bulgaria and Romania came in 2007, Slovenia and Hungary in 2004.

35. On 1 January 2023, Croatia:

- A)** left the Schengen area.
- B)** adopted the euro and joined the Schengen border-free area.
- C)** adopted the Swiss franc.
- D)** withdrew from the European Union.
- E)** changed its currency to the kuna.

Answer: B

On 1 January 2023 Croatia achieved a double first: it became the twentieth euro-area member and simultaneously joined the Schengen area, retiring its old kuna. Withdrawing from the Union or Schengen, or adopting the kuna or the Swiss franc, inverts the facts.

36. On 1 January 2026 Bulgaria became the 21st country to adopt the euro. Its former currency was:

- A)** the dinar **B)** the koruna **C)** the lev **D)** the forint **E)** the zloty

Answer: C

Bulgaria's former currency was the lev, converted at the fixed rate of 1.95583 lev per euro when it became the twenty-first euro country on 1 January 2026. The dinar belongs to Serbia, the koruna to Czechia, the zloty to Poland and the forint to Hungary.

37. The Schengen area is best described as:

- A)** the collective name for the EU's outermost regions.
- B)** a free-trade zone limited to agriculture.
- C)** a currency union.
- D)** a military defence alliance.
- E)** a group of European countries that have abolished controls at their common internal borders.

Answer: E

Schengen is the zone in which participating countries have abolished controls at their common internal borders; it overlaps with, but is not, the EU (Switzerland, Norway, Iceland and Liechtenstein take part, while EU members Ireland and Cyprus do not). It is not a military alliance, a currency union or an agricultural scheme.

38. The European Union as an organisation was awarded the Nobel Peace Prize in:

- A)** 1993 **B)** 1999 **C)** 2007 **D)** 2012 **E)** 2020

Answer: D

The European Union received the Nobel Peace Prize in 2012 for over six decades of advancing peace, reconciliation, democracy and human rights in Europe. None of the other years listed carries this award.

39. Which institution of the European Union is directly elected by the citizens?

- A)** The Court of Auditors.
- B)** The European Parliament.
- C)** The European Council.
- D)** The Council of the EU.
- E)** The European Commission.

Answer: B

The European Parliament is the only EU institution elected directly by citizens, every five years by universal suffrage since 1979; the June 2024 election seated 720 MEPs. The Commission, the two Councils and the Courts are all appointed bodies.

40. Which of the following EU member states does not use the euro?

- A)** Denmark **B)** Greece **C)** Spain **D)** France **E)** Portugal

Answer: A

Denmark keeps its krone under a negotiated opt-out from the euro (the krone is nevertheless pegged through ERM II). France, Spain, Portugal and Greece are all euro users from the currency's start.

Section D - Italian History (Q41-54)

41. The Kingdom of Italy was proclaimed in:

- A)** 1848 **B)** 1859 **C)** 1861 **D)** 1866 **E)** 1870

Answer: C

The Kingdom of Italy was proclaimed on 17 March 1861 in Turin, with Vittorio Emanuele II as first king. Veneto arrived in 1866 and Rome in 1870, so those later dates mark enlargements, not the proclamation.

42. The 'Expedition of the Thousand' (1860), which conquered the Kingdom of the Two Sicilies, was led by:

- A)** Giuseppe Mazzini
B) Vittorio Emanuele II
C) Camillo Benso di Cavour
D) Vincenzo Gioberti
E) Giuseppe Garibaldi

Answer: E

Giuseppe Garibaldi led the Thousand, landing at Marsala in 1860, conquering Sicily and Naples and handing the south to Vittorio Emanuele II. Cavour directed diplomacy from the north, Mazzini inspired from exile, and Gioberti was a pamphleteer, none of them commanding the expedition.

43. The prime minister of Piedmont who engineered France's support and diplomatic cover for unification was:

- A)** Camillo Benso di Cavour
B) Giuseppe Garibaldi
C) Giuseppe Mazzini
D) Luigi Einaudi
E) Francesco Crispi

Answer: A

Camillo Benso di Cavour, Piedmont's prime minister, secured Napoleon III's support at Plombières (1858), fought Austria in 1859 and engineered the plebiscites of 1860 that attached central Italy. Einaudi was a later Republic's president, Crispi a later premier.

44. Giuseppe Mazzini is best remembered as:
- A) the first king of united Italy.
 - B) the ideologue of the Fascist party.
 - C) the inventor of the electric telegraph.
 - D) founder of the republican movement 'Young Italy' (Giovine Italia).
 - E) Napoleon's viceroy of Italy.

Answer: D

Mazzini founded Giovine Italia (Young Italy) in 1831 and gave the Risorgimento its republican, unitary ideal, the soul of the triad completed by Cavour's statecraft and Garibaldi's sword. He was no king, viceroy, scientist or fascist ideologue.

45. Rome became the capital of Italy in:
- A) 1861 B) 1866 C) 1870 D) 1871 E) 1878

Answer: D

Troops breached Porta Pia on 20 September 1870, ending papal temporal rule, and Rome formally became the capital in 1871, after Turin (1861 to 1865) and Florence (1865 to 1871). The 1870 date marks the capture, 1871 the transfer of the capital.

46. Italy entered the First World War in 1915 on the side of:
- A) the Central Powers (Germany and Austria-Hungary).
 - B) the Triple Entente (France, Britain, Russia).
 - C) neither — it remained neutral until 1918.
 - D) the Ottoman Empire.
 - E) Spain.

Answer: B

In May 1915 Italy entered the war on the Triple Entente side (France, Britain, Russia), despite its 1882 Triple Alliance with Germany and Austria-Hungary, after the secret Treaty of London promised Austro-Hungarian lands. Neutrality had covered 1914 only, not the whole war.

47. The disastrous Italian defeat of October 1917 — followed by recovery and victory at Vittorio Veneto in 1918 — was the battle of:
- A) Marengo B) El Alamein C) Custoza D) Lissa E) Caporetto

Answer: E

Caporetto, in October 1917, was the German-Austrian breakthrough that shattered the Italian front and forced the retreat to the Piave, avenged a year later by victory at Vittorio Veneto. Custoza and Lissa were the defeats of 1866, Marengo a Napoleonic battle of 1800, El Alamein a 1942 desert battle.

48. The 'March on Rome' (October 1922) resulted in:
- A) Mussolini being appointed Prime Minister by King Vittorio Emanuele III.
 - B) the signing of the Lateran Pacts.
 - C) the proclamation of the Republic.
 - D) the abdication of the king.
 - E) Italy's entry into the First World War.

Answer: A

The March on Rome of October 1922 ended when Vittorio Emanuele III declined to sign the state of siege and appointed Mussolini prime minister, so power passed through a constitutional door. No king abdicated, the Republic came in 1946, and the Lateran Pacts are 1929.

49. The Lateran Pacts of 1929 between Italy and the Holy See:
- A) ended Italy's membership of the League of Nations.
 - B) established the sovereign Vatican City State, ending the 'Roman Question'.
 - C) transferred Venice to Austria.
 - D) created the region of Trentino-Alto Adige.
 - E) nationalised all Church property.

Answer: B

The 1929 Lateran Pacts (a political treaty, a concordat and a financial convention) created the sovereign Vatican City State and settled the Roman Question open since 1870. They did not touch Venice, the League of Nations or Trentino-Alto Adige, nor did they nationalise Church property.

50. Italy's Liberation Day, 25 April, marks:
- A) the 1969 Piazza Fontana bombing.
 - B) the 1946 institutional referendum.
 - C) the 1945 general insurrection and liberation of Milan and Turin from Nazi-fascist occupation.
 - D) the fall of the Berlin Wall.
 - E) the end of the First World War.

Answer: C

On 25 April 1945 the National Liberation Committee called a general insurrection as the German retreat began, and Milan and Turin freed themselves; the date is kept as Liberation Day. The institutional referendum belongs to 2 June 1946, and the other events are unrelated to the date.

51. On 2 June 1946, Italians voted:
- A) to restore the monarchy by a narrow margin.
 - B) to adopt the single European currency.
 - C) to join the North Atlantic Treaty Organization.
 - D) for the first President of the Republic directly.
 - E) in a referendum that founded the Republic and elected the Constituent Assembly — women voting nationally for the first time.

Answer: E

The vote of 2 June 1946 chose a Republic over the monarchy (about 54 to 46 per cent) and elected the Constituent Assembly, the first national election in which women voted. The first President, De Nicola, was chosen by Parliament, not directly, and NATO dates from 1949.

52. Italy's post-war 'economic miracle' — rapid industrial growth, the Fiat 500, mass internal migration — occurred chiefly in the:
- A) first decade of the 2000s.
 - B) 1930s.
 - C) late 1940s.
 - D) 1950s and 1960s.
 - E) 1980s.

Answer: D

The economic miracle ran from the late 1950s into the early 1960s (peak roughly 1958 to 1963): booming industry and exports, the Fiat 500 and mass migration from south to north. The Fascist 1930s, the reconstruction years and the 2000s all miss it.

53. The kidnapping and murder of former Prime Minister Aldo Moro by the Red Brigades occurred in:

- A)** 1969 **B)** 1974 **C)** 1978 **D)** 1980 **E)** 1992

Answer: C

The Red Brigades kidnapped Aldo Moro on 16 March 1978 and murdered him on 9 May 1978 after 55 days in captivity, the climax of the Years of Lead. Piazza Fontana (1969) and Bologna station (1980) bracket but do not date the Moro case.

54. The 'Mani pulite' (Clean Hands) judicial investigations opened in 1992 and:

- A)** investigated doping in professional football.
B) exposed systemic bribery ('Tangentopoli'), dissolving the party system of the 'First Republic'.
C) nationalised the major banks.
D) prosecuted Nazi war criminals.
E) rewrote the constitution's labour articles.

Answer: B

Growing out of a Milan bribery inquiry in 1992, Clean Hands exposed Tangentopoli, the systemic bribery that financed the governing parties, and swept away Christian Democracy and the Socialists, ending the First Republic. War criminals, banks, the constitution's labour articles and football doping were not its targets.

Section E - Art, Architecture & Heritage (Q55-66)

55. Leonardo da Vinci's Last Supper is a mural painted in:

- A)** the refectory of Santa Maria delle Grazie, Milan.
B) the Louvre, Paris.
C) the Uffizi Gallery, Florence.
D) St Peter's Basilica, Rome.
E) the Sistine Chapel, Vatican City.

Answer: A

Leonardo painted the Last Supper (about 1495 to 1498) on the refectory wall of Santa Maria delle Grazie in Milan, experimenting with tempera on dry wall, which explains its famous flaking. The Louvre keeps his Mona Lisa, not this mural.

56. Michelangelo's David (1501-04) stands today in:

- A)** the Louvre, Paris.
B) the Boboli Gardens.
C) the Galleria dell'Accademia, Florence.
D) the Uffizi Gallery.
E) the Vatican Museums.

Answer: C

The David (1501 to 1504) has stood in the Galleria dell'Accademia in Florence since 1873; a replica marks its original spot outside the Palazzo Vecchio. The Uffizi is Botticelli's home ground, not the David's.

57. Michelangelo's Pietà (1499), the only work he ever signed, is in:

- A) St Peter's Basilica, Rome.
- B) the Galleria dell'Accademia, Florence.
- C) the Prado, Madrid.
- D) Milan Cathedral.
- E) Florence Cathedral.

Answer: A

Michelangelo carved the Roman Pietà around his twenty-fourth year and signed it, his only signed work; it rests in St Peter's Basilica behind glass installed after the 1972 hammer attack. Florence's Accademia houses the David, a different masterpiece.

58. Raphael's The School of Athens is frescoed in:

- A) the Sistine Chapel.
- B) the Stanza della Segnatura of the Vatican Apostolic Palace.
- C) the Uffizi Gallery, Florence.
- D) the Scrovegni Chapel, Padua.
- E) the Doge's Palace, Venice.

Answer: B

The School of Athens (1509 to 1511), painted for Pope Julius II, adorns the Stanza della Segnatura of the Vatican Apostolic Palace, with Plato and Aristotle at the centre. The Sistine Chapel is Michelangelo's domain, the Scrovegni Chapel in Padua Giotto's.

59. Botticelli's The Birth of Venus hangs in:

- A) the Vatican Pinacoteca.
- B) the National Gallery, London.
- C) the Galleria Borghese, Rome.
- D) the Uffizi Gallery, Florence.
- E) the Louvre, Paris.

Answer: D

Botticelli's Birth of Venus (about 1485) hangs in the Uffizi in Florence beside his Primavera, twin icons of Medici-era Neo-Platonism. The London, Paris and Rome collections named hold other works.

60. The ceiling of the Sistine Chapel was painted by Michelangelo between:

- A) 1450 and 1455. B) 1490 and 1495. C) 1508 and 1512. D) 1536 and 1541. E) 1575 and 1580.

Answer: C

The Sistine ceiling with its nine Genesis scenes was painted between 1508 and 1512; the altar-wall Last Judgement came a quarter-century later, 1536 to 1541. The two date-pairs must not be swapped.

61. Filippo Brunelleschi's masterpiece of engineering is:
- A) St Mark's Campanile, Venice.
 - B) the Baths of Caracalla.
 - C) the dome of St Paul's Cathedral, London.
 - D) the Leaning Tower of Pisa.
 - E) the double-shell dome of Florence Cathedral (Santa Maria del Fiore).

Answer: E

Brunelleschi raised the double-shell dome of Florence Cathedral, completed by 1436 without traditional wooden centring and still the largest masonry dome, and he also codified linear perspective. St Paul's dome in London is Wren's, and the Leaning Tower is Pisa's Romanesque bell tower.

62. The Pantheon in Rome — still the world's largest unreinforced concrete dome — was essentially completed under the emperor:

A) Hadrian B) Nero C) Constantine D) Augustus E) Trajan

Answer: A

The present Pantheon was rebuilt under Hadrian around 126 AD; only Agrippa's name survives on the façade from the 27 BC original, hence its inscription. Augustus, Trajan and Nero belong to other phases, and Constantine's building came later.

63. The Colosseum — properly the Flavian Amphitheatre — was inaugurated in:

A) 44 BC B) 72 BC C) 80 AD D) 315 AD E) 500 AD

Answer: C

Begun under Vespasian around 70 to 72 AD, the Flavian Amphitheatre was inaugurated by Titus in 80 AD with a hundred days of games. Dates BC predate the Empire, and 315 AD suits the nearby Arch of Constantine era instead.

64. Caravaggio's revolutionary painting style is characterised by:

- A) pastel colours and loose impressionist brushwork.
- B) geometric cubist fragmentation.
- C) serene neoclassical balance.
- D) gold-ground Byzantine flatness.
- E) dramatic contrasts of light and dark with unsparing realism.

Answer: E

Caravaggio's revolution is tenebrism: violent contrasts of light and dark with unsparing realism, saints with dirty feet, as in the Calling of Saint Matthew in Rome's San Luigi dei Francesi. Cubism and impressionism postdate him, neoclassicism opposes his idiom, and gold-ground flatness is medieval.

65. Gian Lorenzo Bernini, the leading sculptor-architect of the Roman Baroque, created:

- A) the dome of Florence Cathedral.
- B) the colonnade of St Peter's Square and The Ecstasy of Saint Teresa.
- C) the David in the Accademia.
- D) the frescoes of the Scrovegni Chapel.
- E) The Birth of Venus.

Answer: B

Bernini shaped Baroque Rome: the colonnade embracing St Peter's Square (1656 to 1667) and the Ecstasy of Saint Teresa (1645 to 1652) in Santa Maria della Vittoria. The Florence dome is Brunelleschi's, the Accademia David Michelangelo's, and the Padua frescoes Giotto's.

66. The archaeological site of Pompeii was buried — and thus preserved — by:

- A)** medieval urban renewal.
- B)** Allied bombing in 1943.
- C)** the 1908 Messina earthquake.
- D)** the eruption of Vesuvius in 79 AD.
- E)** flooding of the Tiber in antiquity.

Answer: D

Vesuvius erupted in 79 AD and buried Pompeii under ash and pumice, sealing the Roman city until excavation; Pliny the Younger's letters describe the catastrophe. Earthquakes and wartime bombing damaged the site but never produced its preservation.

Section F - Literature, Philosophy & Ideas (Q67-80)

67. Dante's Divine Comedy:

- A)** runs to 100 cantos across three canticles — Inferno, Purgatorio, Paradiso — in vernacular Tuscan.
- B)** was composed after the Council of Trent.
- C)** is a collection of 100 sonnets.
- D)** is written throughout in Latin.
- E)** avoids any religious or political content.

Answer: A

The Comedy counts 100 cantos in three canticles (Inferno, Purgatorio, Paradiso), composed around 1308 to 1320 in terza rima in the Tuscan vernacular, the choice that made Dante the father of the Italian language. Writing in Latin would contradict its founding act, and the Council of Trent came two centuries later.

68. Boccaccio's Decameron is framed as:

- A)** a single narrator's deathbed confession.
- B)** a guided dream-tour of the afterlife.
- C)** an exchange of letters between two lovers.
- D)** ten young Florentines telling ten stories each over ten days while plague ravages the city.
- E)** a debate among seven saints.

Answer: D

Boccaccio's frame: ten young Florentines shelter from the 1348 plague in a villa and tell ten stories each over ten days, one hundred novelle in all. The dream-tour of the afterlife is Dante's poem, and single narrators or exchanged letters fit neither the Decameron's structure.

69. Petrarch, author of the Canzoniere addressed to Laura, is traditionally called:

- A)** the father of scientific history.
- B)** the father of Humanism.
- C)** the poet of the Risorgimento.
- D)** the father of experimental medicine.
- E)** the father of Italian opera.

Answer: B

Petrarch is the father of Humanism: his rediscovery of classical texts and the inward, self-scrutinising lyric of the Canzoniere (addressed to Laura, the perfected sonnet) opened the Renaissance; he was crowned poet laureate in Rome in 1341.

70. The Prince (1513) — which argues it is safer for a ruler to be feared than loved — was written by:

- A)** Dante Alighieri
- B)** Erasmus of Rotterdam
- C)** Thomas More
- D)** Baldassare Castiglione
- E)** Niccolò Machiavelli

Answer: E

Machiavelli wrote The Prince in 1513, the founding text of political realism, counselling that a ruler is safer being feared than loved. The Courtier is Castiglione's book, Utopia is More's, and Erasmus wrote satire rather than statecraft.

71. Alessandro Manzoni's novel The Betrothed (I promessi sposi) is set:

- A)** in Fascist-era Rome.
- B)** in seventeenth-century Lombardy, amid war, famine and the plague of Milan.
- C)** in the decades of the Risorgimento.
- D)** during the Roman Republic of 1849.
- E)** in medieval Sicily.

Answer: B

The Betrothed (definitive edition 1840) unfolds in seventeenth-century Lombardy under Spanish rule: Renzo and Lucia, war, bread riots and the plague of Milan of 1630, which Manzoni also dissected in The Column of Infamy. It is a historical novel set two centuries before its writing.

72. The Name of the Rose (1980), a murder mystery set in a fourteenth-century Italian monastery, was written by:

- A)** Elsa Morante **B)** Primo Levi **C)** Grazia Deledda **D)** Umberto Eco **E)** Italo Calvino

Answer: D

Umberto Eco, the Bologna semiotician, wrote The Name of the Rose (1980), the fourteenth-century monastery murder mystery whose detective friar William of Baskerville nods to Sherlock Holmes (filmed with Sean Connery). Levi, Calvino, Morante and Deledda wrote other kinds of fiction.

73. The 1934 Nobel Prize in Literature went to the Italian dramatist of Six Characters in Search of an Author:

- A)** Dario Fo
- B)** Alberto Moravia
- C)** Luigi Pirandello
- D)** Gabriele D'Annunzio
- E)** Eugenio Montale

Answer: C

Luigi Pirandello took the 1934 Nobel in Literature for his theatre of the unstable self, of which Six Characters in Search of an Author is the emblem. Italy's other literature Nobels are Deledda 1926, Quasimodo 1959, Montale 1975 and Fo 1997.

74. If This Is a Man (Se questo è un uomo), the testament of Auschwitz by a deported Jewish chemist, was written by:

- A)** Primo Levi
- B)** Curzio Malaparte
- C)** Giorgio Bassani
- D)** Natalia Ginzburg
- E)** Cesare Pavese

Answer: A

Primo Levi, a Jewish chemist from Turin deported to Auschwitz-Monowitz, wrote If This Is a Man, the survivor's testament whose prefacing poem begs the reader to consider whether this is a man; The Truce followed it.

75. Invisible Cities — in which Marco Polo describes impossible cities to Kublai Khan — is by:

- A)** Umberto Eco
- B)** Luigi Pirandello
- C)** Giuseppe Tomasi di Lampedusa
- D)** Federigo Tozzi
- E)** Italo Calvino

Answer: E

Invisible Cities (1972) is Italo Calvino's catalogue of fifty-five impossible cities that Marco Polo describes to Kublai Khan, each in the end a reflection of Venice. Lampedusa's classic is The Leopard, and Eco wrote the medieval whodunit instead.

76. The Neapolitan Novels, beginning with My Brilliant Friend, are published under the pseudonym:

- A)** Oriana Fallaci
- B)** Dacia Maraini
- C)** Elena Ferrante
- D)** Susanna Tamaro
- E)** Natalia Ginzburg

Answer: C

The Neapolitan Novels, from My Brilliant Friend onward, tracing Lila and Lenù from 1950s Naples, are published under the pseudonym Elena Ferrante, whose real identity has never been confirmed. Fallaci, Ginzburg, Tamaro and Maraini are named, identifiable writers.

77. Plato's allegory of the cave appears in:

- A)** Virgil's Aeneid.
- B)** The Symposium.
- C)** The Phaedo.
- D)** The Nicomachean Ethics.
- E)** The Republic (Book VII).

Answer: E

The allegory of the cave occupies Book VII of Plato's Republic: prisoners take shadows for reality until one climbs toward the sun, Plato's image of education as ascent to the Forms. The Symposium treats love, the Phaedo the soul, and the Nicomachean Ethics is Aristotle's work.

78. Galileo's Dialogue Concerning the Two Chief World Systems (1632) brought him:

- A)** exile to England, where the book was burned.
- B)** a papal pension and a cardinal's hat.
- C)** the discovery of Jupiter's moons.
- D)** trial and condemnation in 1633 for defending Copernican heliocentrism.
- E)** the abolition of the Roman Inquisition.

Answer: D

Publishing the Dialogue in 1632 cost Galileo trial before the Roman Inquisition and condemnation in 1633, with abjuration and house arrest, during which he wrote the Discourses on Two New Sciences. Jupiter's moons had been discovered back in 1610, and the Church's rehabilitation came only in 1992.

79. 'Man is born free, and everywhere he is in chains' opens:

- A)** Rousseau's The Social Contract.
- B)** Kant's Critique of Pure Reason.
- C)** Locke's Two Treatises of Government.
- D)** Hobbes's Leviathan.
- E)** Voltaire's Candide.

Answer: A

The sentence opens Rousseau's Social Contract of 1762, whose standard of legitimacy, the general will, electrified the French Revolution and modern democratic theory. Hobbes opens from the war of all against all, a different first move and a different politics.

80. One Hundred Years of Solitude, the masterpiece of Latin-American magical realism, is by:

- A)** Isabel Allende
- B)** Gabriel García Márquez
- C)** Jorge Luis Borges
- D)** Julio Cortázar
- E)** Mario Vargas Llosa

Answer: B

Gabriel García Márquez published One Hundred Years of Solitude in 1967, the century-spanning saga of the Buendía family in Macondo and the emblem of magical realism; he took the Nobel in 1982. Borges wrote compressed fictions, Vargas Llosa and Cortázar other kinds of novels, and Allende published later.

Section G - Science, Technology & Society (Q81-90)

81. Guglielmo Marconi shared the 1909 Nobel Prize in Physics for:

- A)** wireless telegraphy — the basis of radio.
- B)** the transmission of speech by telephone.
- C)** the first controlled nuclear fission.
- D)** the discovery of the transistor.
- E)** the invention of the electric battery.

Answer: A

Marconi shared the 1909 Physics Nobel with Karl Ferdinand Braun for wireless telegraphy, the technological root of radio. The electric battery is Volta's (1800), the telephone Bell's, the transistor Bell Labs' in 1947, and fission was detected in 1938.

82. Alessandro Volta (1745–1827) is celebrated for:
- A) devising the periodic table.
 - B) discovering X-rays.
 - C) discovering the planet Neptune.
 - D) inventing the electric battery — the 'voltaic pile' — in 1800.
 - E) the first transatlantic radio signal.

Answer: D

Volta built the voltaic pile in 1800, the first electric battery, coaxing steady current from stacked copper and zinc discs; the volt honours his name and Napoleon honoured the man. Neptune, X-rays and the periodic table belong to other scientists, and the transatlantic radio signal is Marconi's feat.

83. Enrico Fermi's team achieved the first controlled nuclear chain reaction on 2 December 1942 at:
- A) the Oak Ridge complex.
 - B) CERN, Geneva.
 - C) Los Alamos.
 - D) the Via Panisperna institute, Rome.
 - E) the University of Chicago.

Answer: E

Chicago Pile-1, under Fermi's direction, went critical on 2 December 1942 beneath the stands of Stagg Field at the University of Chicago. The Via Panisperna institute in Rome was his earlier laboratory, Los Alamos and Oak Ridge later Manhattan Project sites, and CERN is a peacetime laboratory.

84. CERN, the European laboratory that runs the Large Hadron Collider, is located near:
- A) Turin. B) Geneva. C) Paris. D) Vienna. E) Munich.

Answer: B

CERN straddles the Franco-Swiss border near Geneva. Founded in 1954, it detected the Higgs boson in 2012 and is where Tim Berners-Lee invented the World Wide Web in 1989.

85. With his improved telescope in 1610, Galileo discovered:
- A) the canals of Mars.
 - B) the planet Neptune.
 - C) the four largest moons of Jupiter.
 - D) the Antarctic ozone hole.
 - E) Halley's Comet.

Answer: C

Galileo's 1610 observations, announced in *Sidereus Nuncius*, revealed the four largest moons of Jupiter (Io, Europa, Ganymede, Callisto), along with lunar mountains and the phases of Venus, evidence that not everything circles the Earth. Neptune awaited 1846, and the Martian canals were a later optical illusion.

86. Avogadro's law states that equal volumes of gases at the same temperature and pressure:
- A) exert no pressure on their containers.
 - B) contain equal numbers of atoms only.
 - C) contain equal numbers of molecules.
 - D) have equal mass.
 - E) must both be noble gases.

Answer: C

Avogadro's law (1811): equal volumes of any gases at the same temperature and pressure contain equal numbers of molecules, whatever their identity or weight; it grounds the mole and Avogadro's number of 6.022×10^{23} . Equal mass fails at once for gases of different molecular weight.

87. Alexander Fleming's 1928 discovery, later mass-produced for D-Day, was:
- A) insulin.
 - B) penicillin.
 - C) the polio vaccine.
 - D) aspirin.
 - E) chloroquine.

Answer: B

Fleming's 1928 observation of *Penicillium* mould destroying staphylococci on a neglected plate gave penicillin; Florey and Chain turned it into a drug in time for the D-Day landings, and the three shared the 1945 Nobel. Aspirin, insulin, the polio vaccine and chloroquine have entirely different origins.

88. The double-helix structure of DNA (1953) was chiefly worked out by:
- A) Watson and Crick, building crucially on Rosalind Franklin's X-ray data.
 - B) Hershey and Chase.
 - C) Linus Pauling working alone.
 - D) Mendel and Darwin together.
 - E) Chargaff and Griffith.

Answer: A

Watson and Crick built the double-helix model at Cambridge in 1953, building crucially on Rosalind Franklin's X-ray diffraction work (Photo 51) at King's College London; the 1962 Nobel honoured Watson, Crick and Wilkins, Franklin having died in 1958. Pauling had proposed a triple helix, wrongly.

89. The 2024 Nobel Prize in Physics was awarded to John Hopfield and Geoffrey Hinton for:
- A) high-temperature superconductivity.
 - B) the invention of the blue LED.
 - C) the discovery of the Higgs boson.
 - D) the detection of gravitational waves.
 - E) foundational discoveries in artificial neural networks and machine learning.

Answer: E

The 2024 Physics Nobel went to John Hopfield and Geoffrey Hinton for foundational discoveries and inventions that enable machine learning with artificial neural networks, physics-trained methods behind today's AI. Gravitational waves, the Higgs and the blue LED were recognised in 2017, 2013 and 2014, and superconductivity honours went to other physicists.

90. The James Webb Space Telescope, launched on 25 December 2021, observes primarily in:

- A)** gamma rays.
- B)** radio waves.
- C)** ultraviolet only.
- D)** the infrared.
- E)** visible light.

Answer: D

The James Webb Space Telescope, launched on 25 December 2021, observes primarily in the infrared, its 6.5-metre gold-coated segmented mirror parked at the Sun-Earth L2 point to catch light from the earliest, red-shifted galaxies. Hubble is the visible-light predecessor; gamma rays, radio and ultraviolet belong to other observatories.

Section H - Current Affairs 2024 (Q91-100)

91. Following the death of Pope Francis in April 2025, the conclave of 8 May 2025 elected:

- A)** Cardinal Matteo Zuppi as John Paul III.
- B)** Cardinal Pietro Parolin as Benedict XVIII.
- C)** Cardinal Luis Tagle as Pius XIII.
- D)** Cardinal Péter Erdő as Paul VII.
- E)** Cardinal Robert Francis Prevost — Leo XIV, the first American pope.

Answer: E

After Pope Francis died on 21 April 2025, the conclave of 8 May 2025 elected Cardinal Robert Francis Prevost: Pope Leo XIV, born in Chicago in 1955, an Augustinian long active in Peru and the first pope from the United States. Parolin, Zuppi, Tagle and Erdő were widely named contenders, not the choice.

92. The 2026 Winter Olympics (6–22 February 2026) were hosted by:

- A)** Milan and Cortina d'Ampezzo.
- B)** Oslo.
- C)** Turin alone.
- D)** Sapporo.
- E)** Innsbruck.

Answer: A

The 2026 Winter Olympics ran 6 to 22 February 2026 at Milan and Cortina d'Ampezzo, Italy's third Winter Games after Cortina 1956 and Turin 2006. Turin alone dates to 2006, and the Scandinavian and Austrian cities listed host no 2026 Games.

93. The 2026 FIFA World Cup — the first 48-team edition, hosted by the USA, Canada and Mexico — was won by:

- A)** Argentina, beating France.
- B)** Portugal, beating England.
- C)** Brazil, beating Germany.
- D)** Spain, beating Argentina 1–0 after extra time.
- E)** France, beating Portugal.

Answer: D

Spain won the first 48-team World Cup by beating Argentina 1–0 after extra time in the final of 19 July 2026 at MetLife Stadium, Ferran Torres scoring in the 106th minute for a second Spanish title after 2010. Argentina's bid for back-to-back crowns failed at the last step, and France, Portugal, Brazil and England fell short of the trophy.

94. After the Paris 2024 Summer Games — the city's third Olympics — the next Summer Olympics will be held in:

- A)** Tokyo in 2030.
- B)** Los Angeles in 2028.
- C)** Rio de Janeiro in 2026.
- D)** Madrid in 2027.
- E)** Berlin in 2029.

Answer: B

Paris 2024 was that city's third Games (after 1900 and 1924); the Summer Olympics next go to Los Angeles in 2028, and then Brisbane in 2032. The other city-year pairings do not exist on the calendar, and the Summer Games keep their even-year, four-year rhythm.

95. The 2025 Nobel Peace Prize was awarded to:

- A)** the World Health Organization.
- B)** Nihon Hidankyo, Japan's atomic-bomb survivors.
- C)** María Corina Machado, Venezuelan opposition leader.
- D)** Volodymyr Zelensky.
- E)** the International Atomic Energy Agency.

Answer: C

The 2025 Nobel Peace Prize went to María Corina Machado, the Venezuelan opposition leader, for her fight for democratic rights; her daughter collected the medal in Oslo while Machado stayed in hiding. Nihon Hidankyo, the Japanese atomic-bomb survivors' group, was the 2024 laureate, and the WHO, the IAEA and Zelensky have not won it.

96. The 2024 Nobel Prize in Literature was awarded to:

- A)** Jon Fosse.
- B)** László Krasznahorkai.
- C)** Annie Ernaux.
- D)** Kazuo Ishiguro.
- E)** Han Kang — the first South Korean laureate.

Answer: E

The 2024 Nobel in Literature went to Han Kang, the first South Korean laureate, author of *The Vegetarian* and *Human Acts*. Ishiguro won in 2017, Ernaux in 2022, Fosse in 2023, and Hungary's László Krasznahorkai followed as the 2025 laureate.

97. Following the November 2024 US election, the president inaugurated on 20 January 2025 was:

- A)** Ron DeSantis.
- B)** Donald Trump — the 47th president, in a second, non-consecutive term.
- C)** Barack Obama, in an unprecedented third term.
- D)** Joe Biden, for a second term.
- E)** Kamala Harris, the first woman president.

Answer: B

Donald Trump, winner of the November 2024 election, was inaugurated on 20 January 2025 as the 47th president, joining Grover Cleveland as the only presidents to serve two non-consecutive terms (he is both the 45th and the 47th). Biden and Harris were the outgoing ticket, and a third Obama term is constitutionally barred.

98. The Catholic Jubilee (Holy Year) that ran through 2025 under the motto 'Pilgrims of Hope' was centred on:

- A)** Mexico City. **B)** Jerusalem. **C)** Lourdes. **D)** Rome. **E)** Assisi.

Answer: D

The 2025 ordinary Jubilee, on the usual twenty-five-year rhythm and under the motto Pilgrims of Hope, was centred on Rome, opened by Francis at St Peter's on 24 December 2024 and closed by Leo XIV on 6 January 2026. Lourdes, Assisi, Jerusalem and Mexico City host no universal Holy Year of 2025.

99. The 2025 Nobel Prize in Physics honoured work on macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit — the foundation of:

- A)** superconducting quantum computers.
B) mRNA vaccines.
C) nuclear fusion reactors.
D) satellite navigation.
E) perovskite solar panels.

Answer: A

Clarke, Devoret and Martinis won the 2025 Physics Nobel for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit, the physics that superconducting qubits inherit and hence the foundation of superconducting quantum computers. Fusion reactors, mRNA vaccines, perovskite panels and satellite navigation are unrelated fields.

100. The 2025 Nobel Prize in Physiology or Medicine was awarded for discoveries concerning:

- A)** messenger-RNA vaccines.
B) the hepatitis C virus.
C) regulatory T cells and the FOXP3 gene.
D) microRNAs.
E) the CRISPR genetic scissors.

Answer: C

The 2025 Medicine Nobel honoured Mary Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries on peripheral immune tolerance: regulatory T cells and the FOXP3 gene, the immune system's brakes, knowledge central to autoimmunity and transplant medicine. MicroRNA took the 2024 prize, mRNA vaccines 2023, CRISPR the 2020 Chemistry prize and hepatitis C the 2020 Medicine prize.